COFFEE SHOP SALES ANALYSIS

1. Project Overview and Objective

This project focuses on analysing coffee shop sales data using **Microsoft Power BI Desktop**. The dataset was cleaned and transformed using Power Query, followed by data modelling and DAX calculations to create meaningful business insights.

The objective of this project is to develop an interactive Power BI dashboard that helps analyse sales performance, product demand, and store performance. The dashboard allows users to monitor revenue trends, identify top-selling products, and understand customer purchasing patterns across different store locations and time periods.

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2. Data Sources

- **Source Description and Timeline:** The dataset contains transactional information about coffee shop purchases, including product category, product type, store location, quantity sold, price, and transaction time.
- **Domain**: Food & Beverage Analytics
- **Source link**: https://mavenanalytics.io/data-playground/coffee-shop-sales

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3. Problem Statement

- To analyse overall sales performance of the coffee shop.
- To identify top-selling product categories and products.
- To evaluate store performance across different locations.
- To analyse customer purchasing patterns by time of day.
- To understand sales trends over time.

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4. Attribute (Column /Features) Details

Attribute Name	Data Type	Description
transaction_id	Integer	Unique identifier for each transaction
transaction_date	Date	Date on which the transaction occurred
transaction_time	Time	Time when the purchase was made
store_location	Text	Location of the coffee shop store
product_category	Text	Category of the product such as coffee or bakery
product_type	Text	Type of product sold

product_detail	Text	Detailed name of the product
transaction_qty	Integer	Quantity of products purchased
unit_price	Decimal Number	Price per unit of product
Total Sales	Decimal Number	Total revenue generated from each transaction
Hour	Whole number	Represents customer purchasing patterns at different hours of the day and identify peak sales periods.
Time_period	Text	Categorizes the transaction time into different periods of the day
Date	Date	Represents the calendar date of each transaction.

5. Tools & Technologies

Power BI: Data cleaning, transformation, modelling, DAX calculations, and interactive dashboard creation.

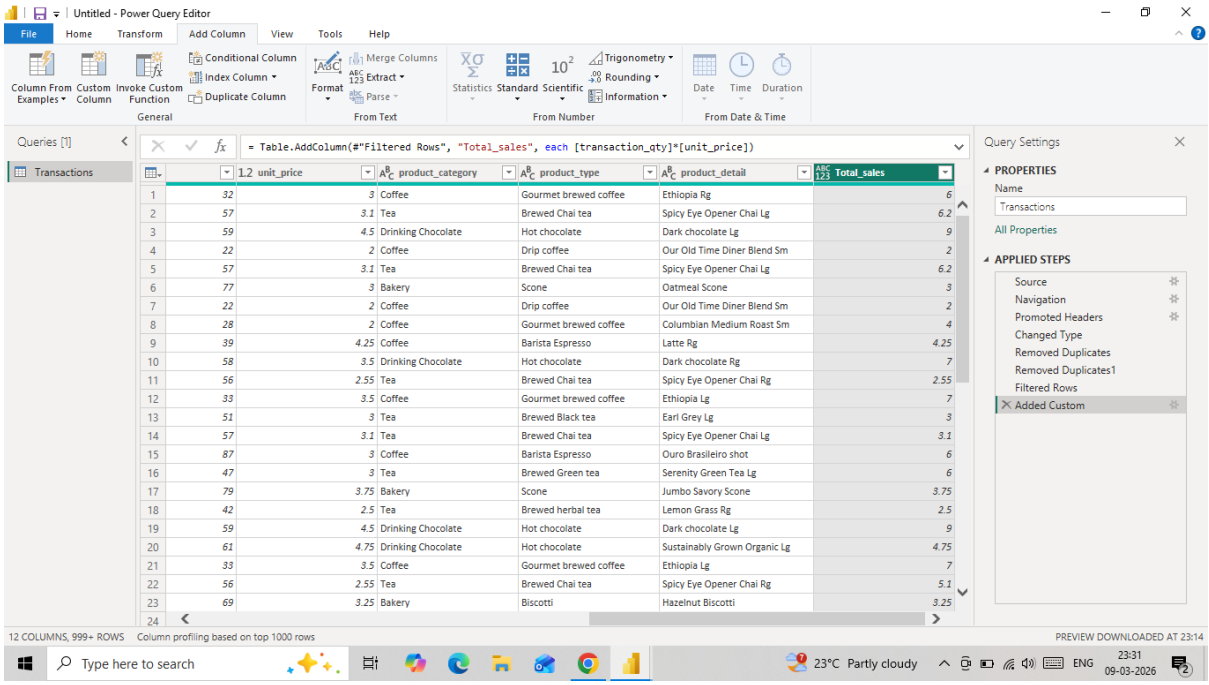
6. Data Pre-Processing (Power Query)

1. Data Cleaning & Transformation:

Removed duplicates, handled missing values, standardized formats, and created calculated fields.

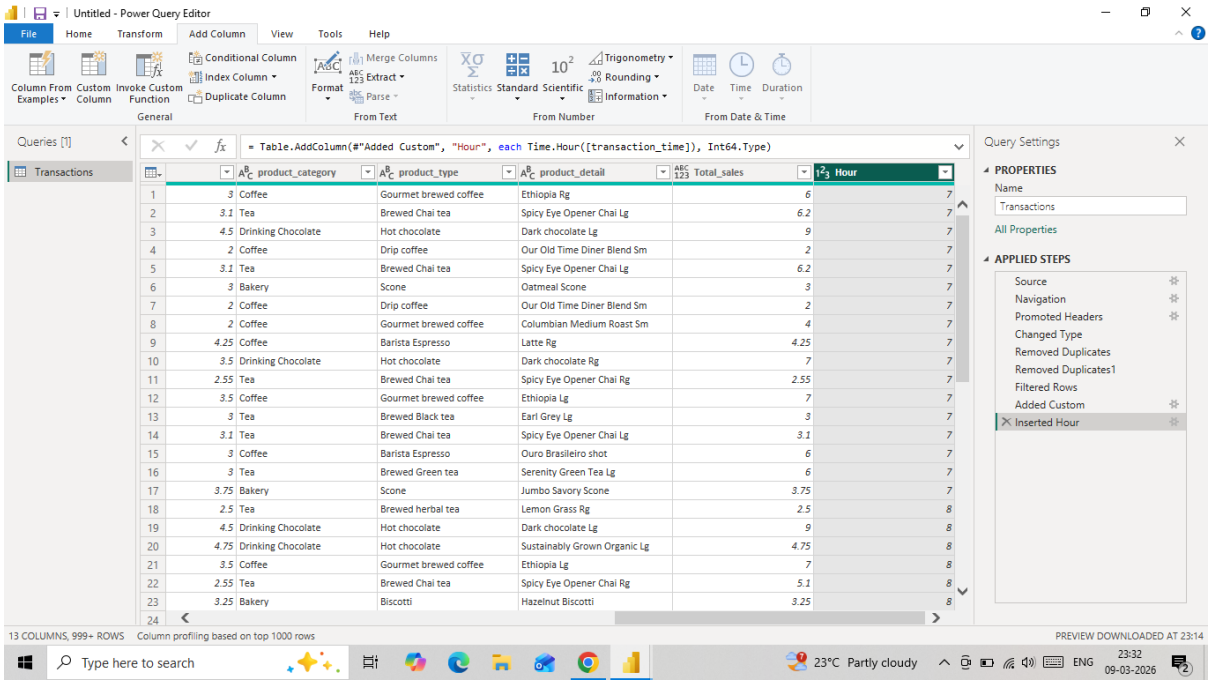
1. Total Sales column was created to calculate revenue per transaction:

Total Sales = [transaction_qty] * [unit_price]



2. We will analyze sales by hour:

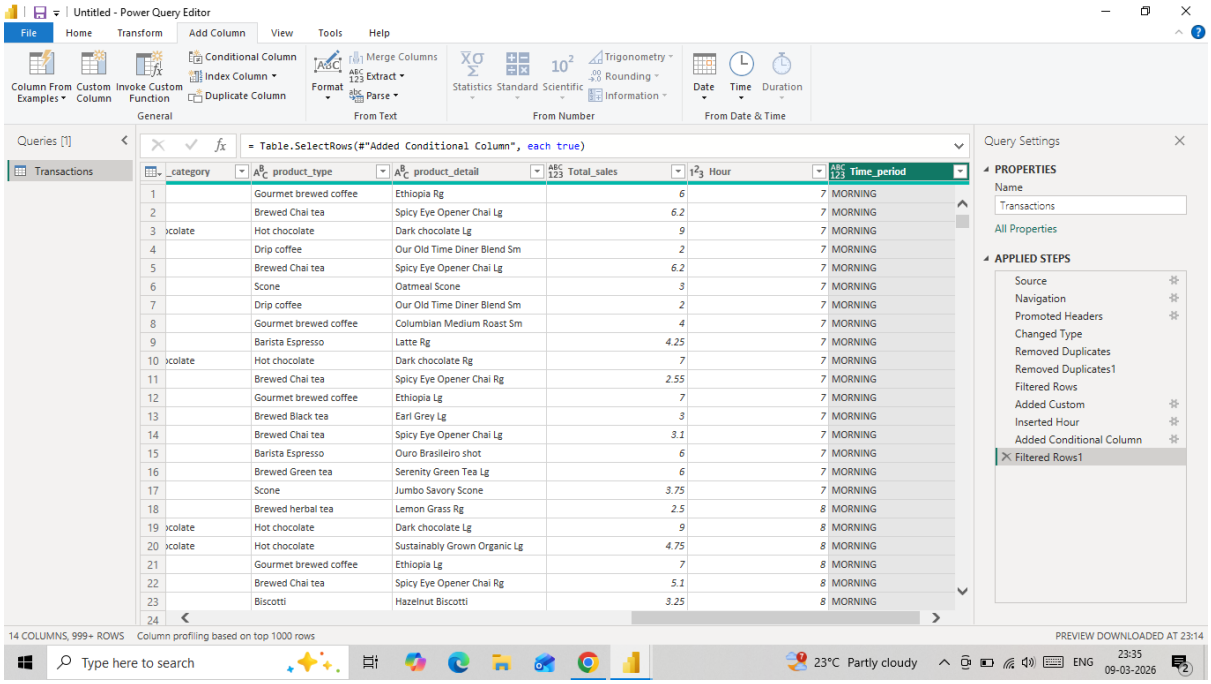
New column: **Hour**→ Add Column → Time → Hour



3. Created Time Category:

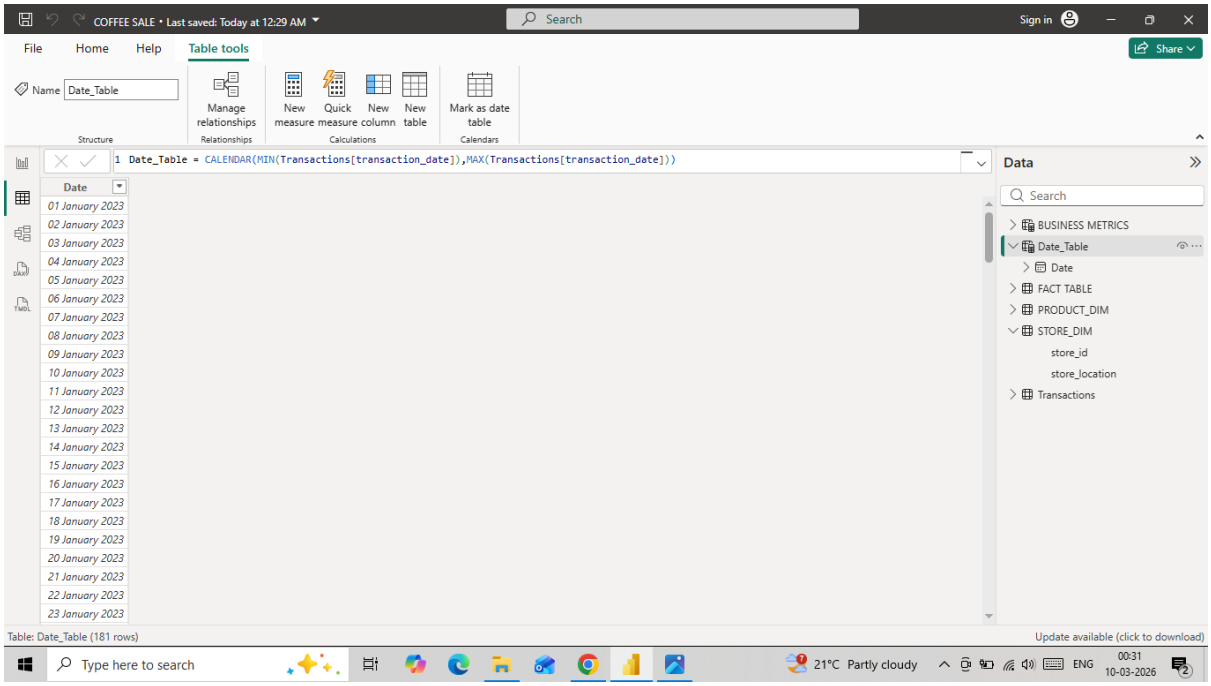
Add a **Conditional Column:** Column Name: **Time Period**

CONDITION	OUTPUT
Hour ≤ 12	Morning
Hour ≤ 17	Afternoon
Hour ≤ 21	Evening
Else	Night

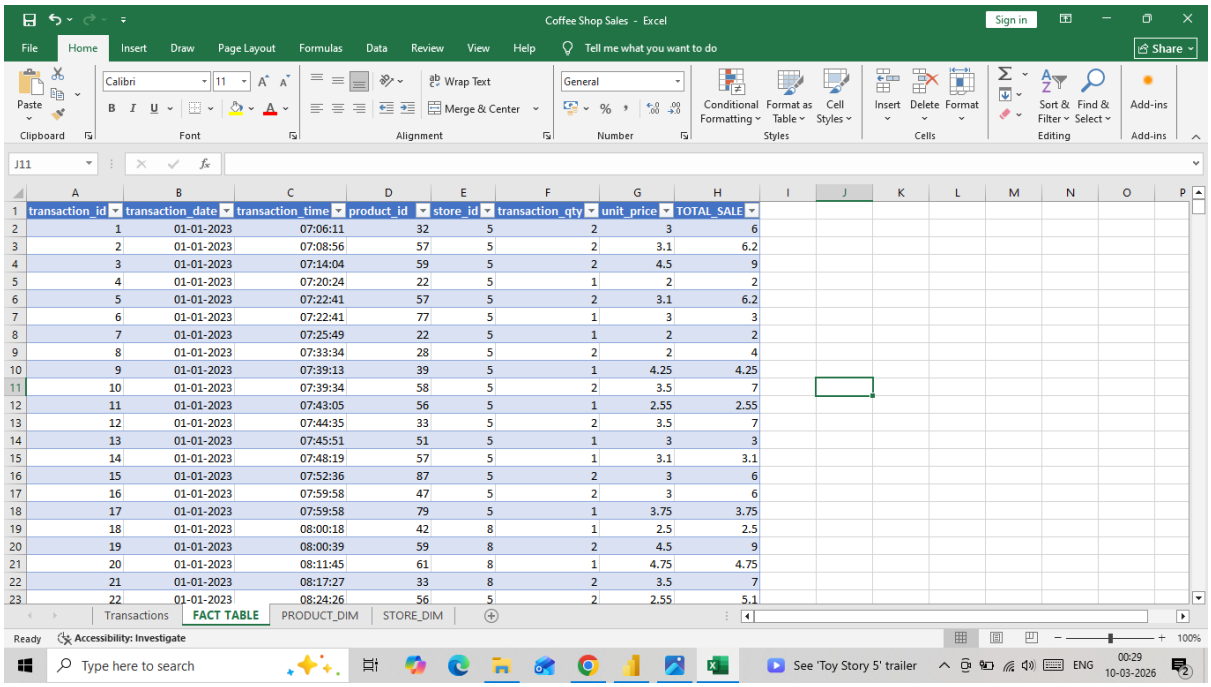


4. Created New Table:

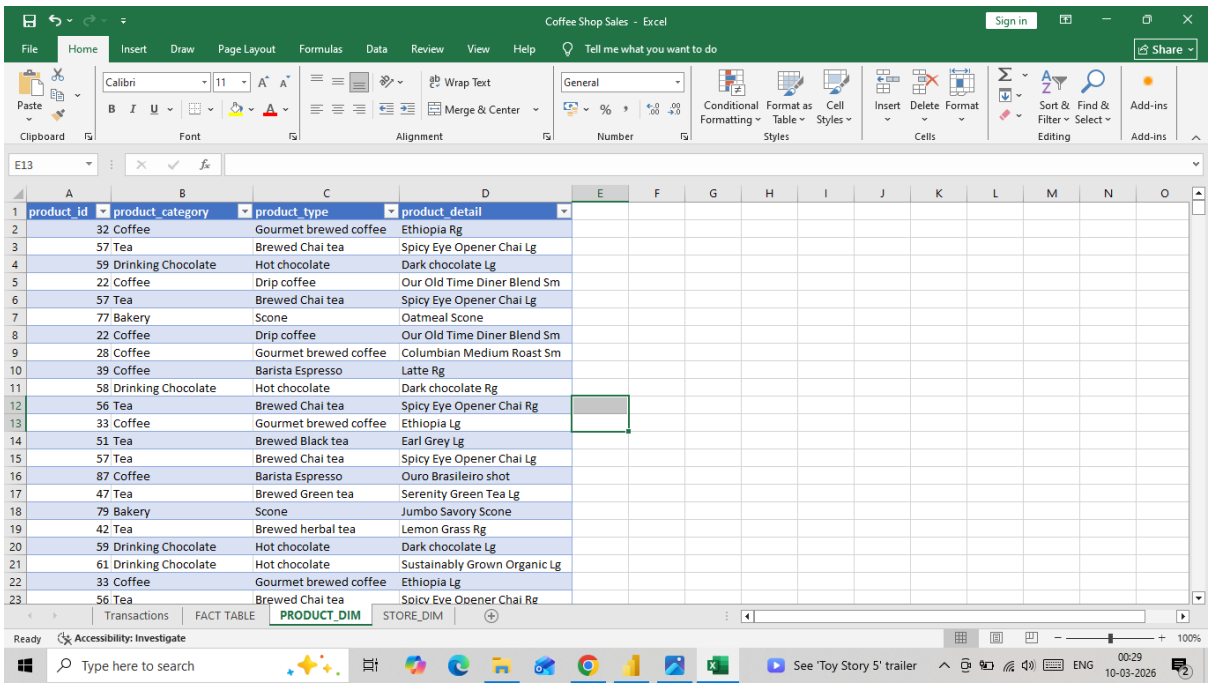
Date_table= CALENDER(MIN(Transaction[DATE]),MAX(Transaction[DATE]))



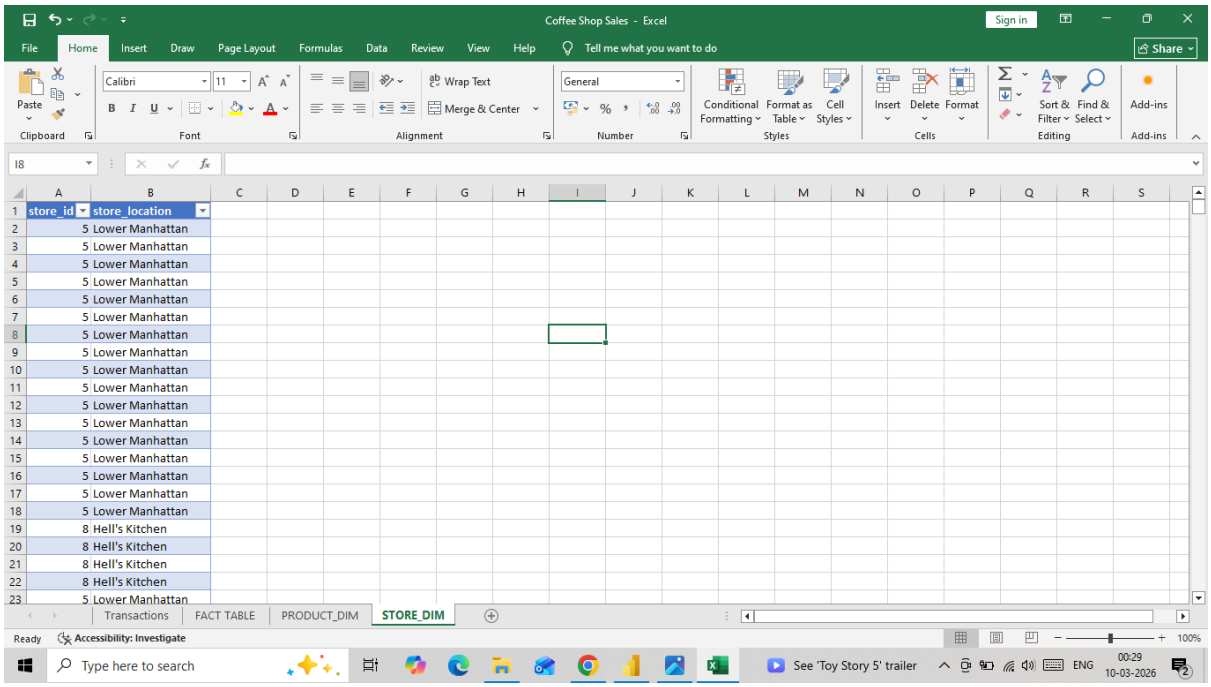
2. Convert the data into Fact and Dimension Table:



transaction_id	transaction_date	transaction_time	product_id	store_id	transaction_qty	unit_price	TOTAL_SALE
1	01-01-2023	07:06:11	32	5	2	3	6
2	01-01-2023	07:08:56	57	5	2	3.1	6.2
3	01-01-2023	07:14:04	59	5	2	4.5	9
4	01-01-2023	07:20:24	22	5	1	2	2
5	01-01-2023	07:22:41	57	5	2	3.1	6.2
6	01-01-2023	07:22:41	77	5	1	3	3
7	01-01-2023	07:25:49	22	5	1	2	2
8	01-01-2023	07:33:34	28	5	2	2	4
9	01-01-2023	07:39:13	39	5	1	4.25	4.25
10	01-01-2023	07:39:34	58	5	2	3.5	7
11	01-01-2023	07:43:05	56	5	1	2.55	2.55
12	01-01-2023	07:44:35	33	5	2	3.5	7
13	01-01-2023	07:45:51	51	5	1	3	3
14	01-01-2023	07:48:19	57	5	1	3.1	3.1
15	01-01-2023	07:52:36	87	5	2	3	6
16	01-01-2023	07:59:58	47	5	2	3	6
17	01-01-2023	07:59:58	79	5	1	3.75	3.75
18	01-01-2023	08:00:18	42	8	1	2.5	2.5
19	01-01-2023	08:00:39	59	8	2	4.5	9
20	01-01-2023	08:11:45	61	8	1	4.75	4.75
21	01-01-2023	08:17:27	33	8	2	3.5	7
22	01-01-2023	08:24:26	56	5	2	2.55	5.1



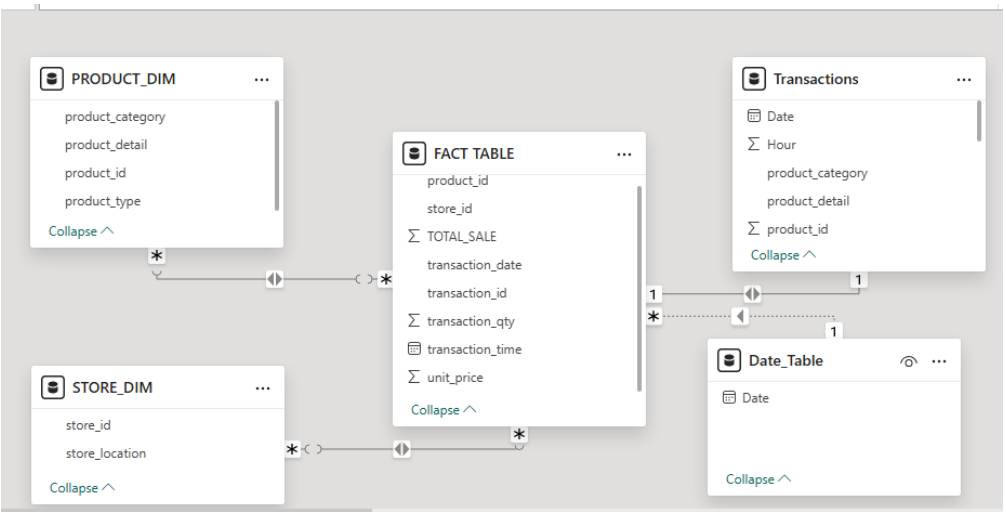
product_id	product_category	product_type	product_detail
32	Coffee	Gourmet brewed coffee	Ethiopia Rg
57	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
59	Drinking Chocolate	Hot chocolate	Dark chocolate Lg
22	Coffee	Drip coffee	Our Old Time Diner Blend Sm
57	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
77	Bakery	Scone	Oatmeal Scone
22	Coffee	Drip coffee	Our Old Time Diner Blend Sm
28	Coffee	Gourmet brewed coffee	Columbian Medium Roast Sm
39	Coffee	Barista Espresso	Latte Rg
58	Drinking Chocolate	Hot chocolate	Dark chocolate Rg
56	Tea	Brewed Chai tea	Spicy Eye Opener Chai Rg
33	Coffee	Gourmet brewed coffee	Ethiopia Lg
51	Tea	Brewed Black tea	Earl Grey Lg
57	Tea	Brewed Chai tea	Spicy Eye Opener Chai Lg
87	Coffee	Barista Espresso	Ouro Brasileiro shot
47	Tea	Brewed Green tea	Serenity Green Tea Lg
79	Bakery	Scone	Jumbo Savory Scone
42	Tea	Brewed herbal tea	Lemon Grass Rg
59	Drinking Chocolate	Hot chocolate	Dark chocolate Lg
61	Drinking Chocolate	Hot chocolate	Sustainably Grown Organic Lg
33	Coffee	Gourmet brewed coffee	Ethiopia Lg
56	Tea	Brewed Chai tea	Spicy Eye Opener Chai Rg



store_id	store_location
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
5	Lower Manhattan
8	Hell's Kitchen
8	Hell's Kitchen
8	Hell's Kitchen
8	Hell's Kitchen
5	Lower Manhattan

7. Data Modelling and DAX(Power BI)

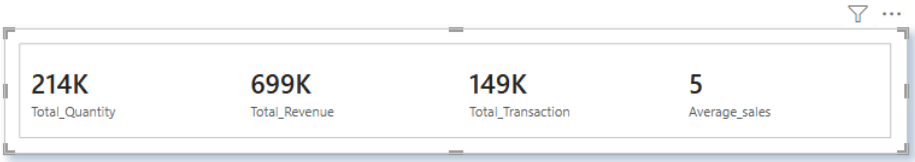
Data Model: Established relationships between tables, defined cardinality, and created lookup tables where necessary.



- The dataset was structured using a **Star Schema Data Model** to improve query performance and simplify reporting. In this structure, a central Fact Table stores the measurable transactional data, while multiple Dimension Tables store descriptive attributes used for filtering and categorizing the data.
- The **FACT_TABLE** contains the primary transactional data related to coffee shop sales. Each record in this table represents a single transaction and includes numerical values that are used for calculations and analysis.
- The **DIMENSION_TABLES** contain descriptive information that helps categorize and filter the data in the fact table. The product_dim table stores product-related information. It helps analyse sales based on product characteristics. The store_dim table contains information about the coffee shop store locations. The date_table is a time dimension created to support time-based analysis the trends by day, month, and year, enabling time-based reporting and trend analysis.

Calculated Columns & DAX Measures: Implemented DAX formulas for key metrics, such as total sales, growth percentages, and cumulative totals**.**

```
Average_sales = AVERAGE('FACT TABLE'[TOTAL_SALE])
Mean = AVERAGE(Transactions[Total_sales])
Median = MEDIAN(Transactions[Total_sales])
Standard_deviation = STDEV.P(Transactions[Total_sales])
Total_quantity = SUM(Transactions[transaction_qty])
Total_revenue = SUM('FACT TABLE'[TOTAL_SALE])
Total_transaction =  COUNT(Transactions[transaction_id])
```



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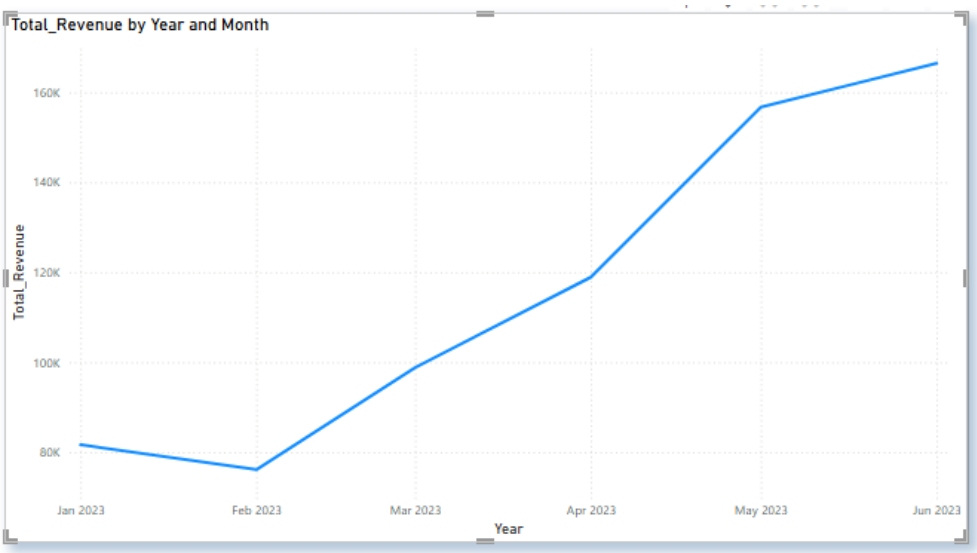
8. Analysis and Visualizations (Power BI)

1. Total Revenue Trend by Year and Month:

Insights

Revenue shows a steady increase from March to June, indicating growing customer demand and improved sales performance during these months. A slight decline in February compared to January suggests a temporary drop in sales before the business started recovering. The highest revenue is recorded in June, suggesting that the coffee shop experienced its peak sales during this period.

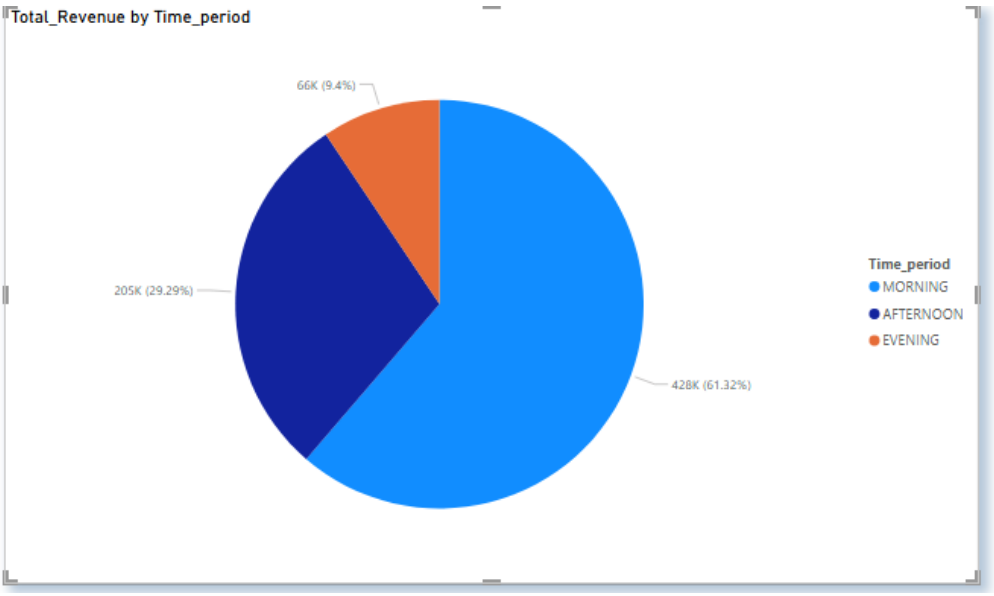


2. Distribution of Total Revenue by Time Period:

Insights

Dominant Revenue Window: The Morning period is the primary driver of sales, contributing a significant 61.32% (428K) of the total revenue. Secondary Peak: The Afternoon session accounts for nearly 29.29% (205K) of sales, representing the second-highest activity level of the day. Evening Decline: There is a noticeable drop in sales during the Evening, which generates the lowest portion of revenue at just 9.4% (66K).

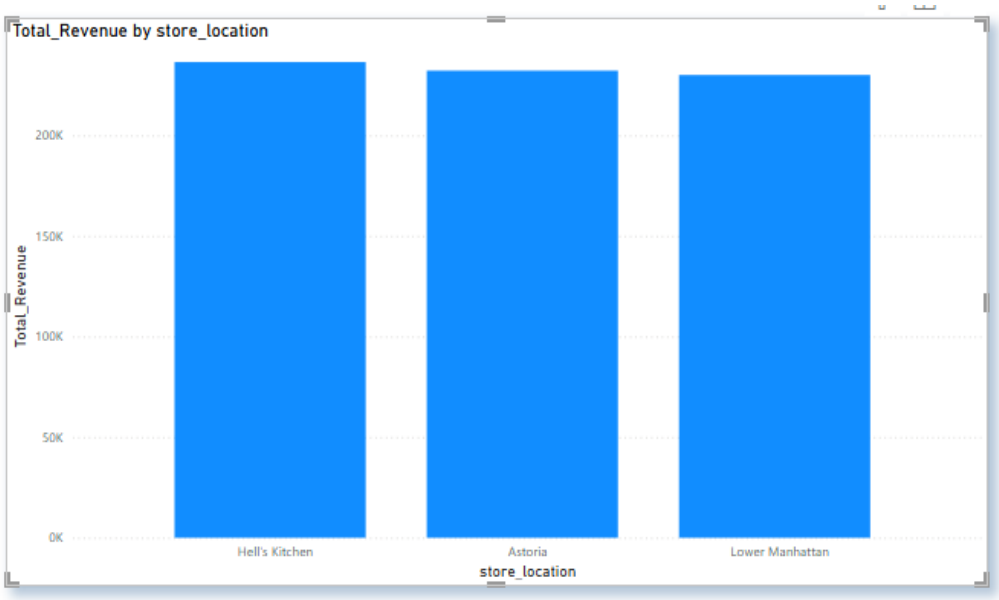
The data suggests that the majority of customer transactions occur before the evening, highlighting a major opportunity for morning-specific marketing or staffing optimization.



3. Revenue Performance by Store Location:

Insights

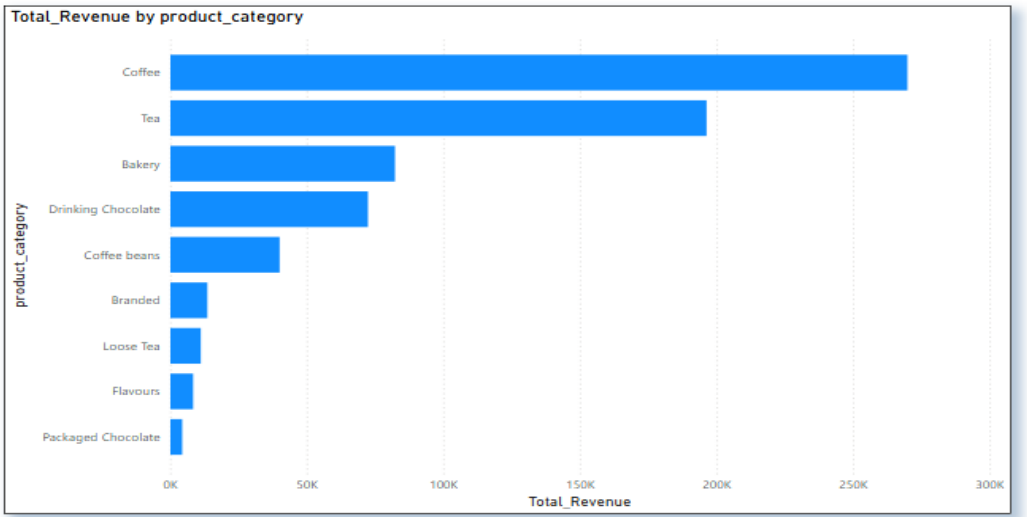
Revenue is remarkably balanced across all locations, with each store contributing roughly equal shares. Hell's Kitchen emerges as the lead location, slightly outperforming the other branches in total earnings. Astoria holds the second position, showing a strong and stable customer base. Lower Manhattan, while competitive, shows the most room for targeted local marketing to bridge the small gap with the top performers.



4. Category Wise Revenue Distribution:

Insights

Coffee is the undisputed primary revenue driver, significantly outpacing all other categories. Tea and Bakery items represent the next most profitable segments, proving essential to the product mix. Drinking Chocolate and Coffee Beans maintain moderate sales, catering to a specific niche of the customer base. Packaged Chocolate and Flavours contribute the least, suggesting these may be impulse buys rather than destination products.



5. Geospatial Distribution of Sales and Transactions:

Insights

The map reveals a bi-coastal operational strategy, with major business hubs established in both the Northwest and Northeast regions of the United States. Both primary locations show similar bubble sizes, indicating a balanced distribution of total revenue and transaction volume across different geographic territories. The data points are concentrated in high-density areas (near Washington and New York/New Jersey), suggesting the business successfully captures metropolitan consumer traffic.

6. Revenue Composition by Specific Product Type:

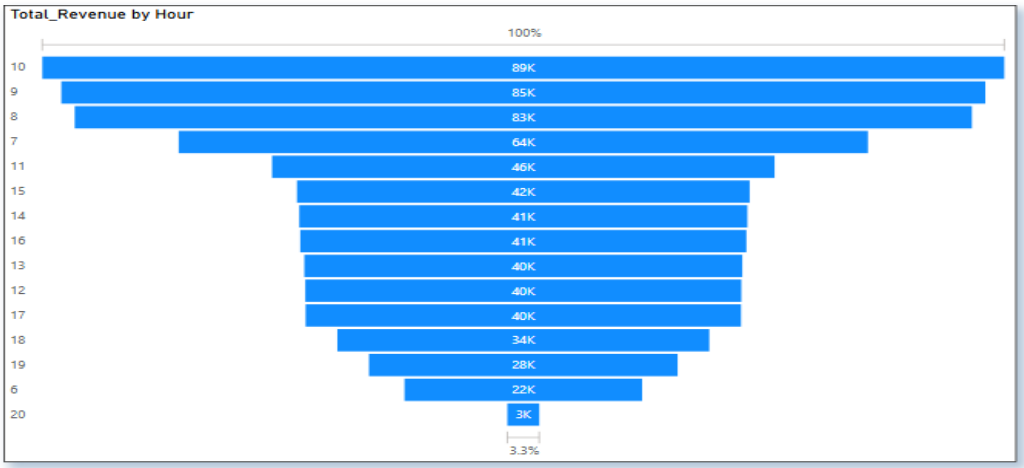
Insights

Barista Espresso and Brewed Chai Tea stand out as the largest contributors to total revenue, indicating strong customer preference for prepared beverages. High-margin items like Gourmet Brewed Coffee and Premium Brewed Coffee occupy significant shares, highlighting the brand's successful positioning in the specialty market.

7. Hourly Revenue Performance Funnel:

Insights

The 10:00 AM hour represents the highest revenue peak at $89K, confirming that the late morning is the most critical period for sales. Revenue builds rapidly starting at 8:00 AM ($83K) and 9:00 AM ($85K), indicating a high volume of breakfast and early-office-hour commuters. After the 11:00 AM mark, revenue drops by nearly 50% and stabilizes around the $40K range throughout the afternoon, showing steady but lower-intensity traffic. Sales reach their lowest point during the 8:00 PM (20:00) hour with only $3K, suggesting that extending operating hours further into the night may not be cost-effective.

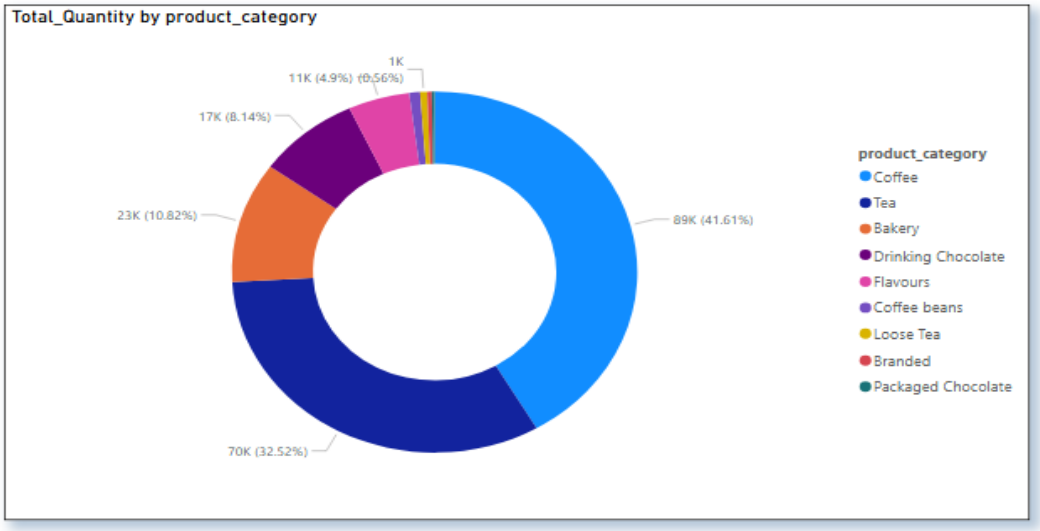


8. Total Quantity Distribution by Product Category:

Insights

Coffee is the highest-volume category, accounting for 41.61% (89K units) of all items sold, cementing its status as the business anchor. Tea follows closely as a major contributor, making up nearly 32.52% (70K units) of the total quantity distributed.

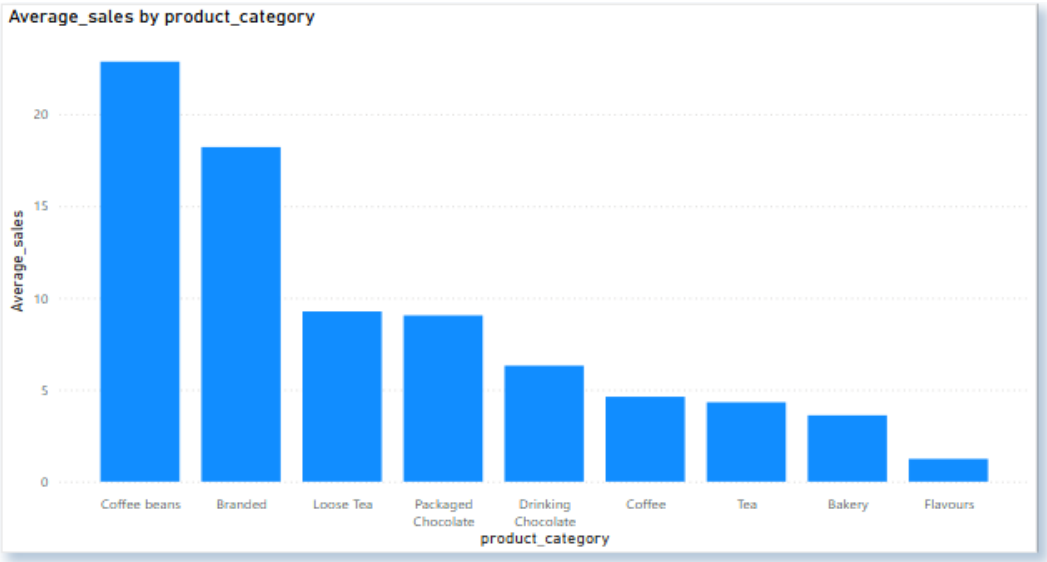
Bakery items represent 10.82% (23K units) of sales, indicating a healthy rate of food-to-beverage pairing among customers.



9. Average Sales Value per Product Category:

Insights

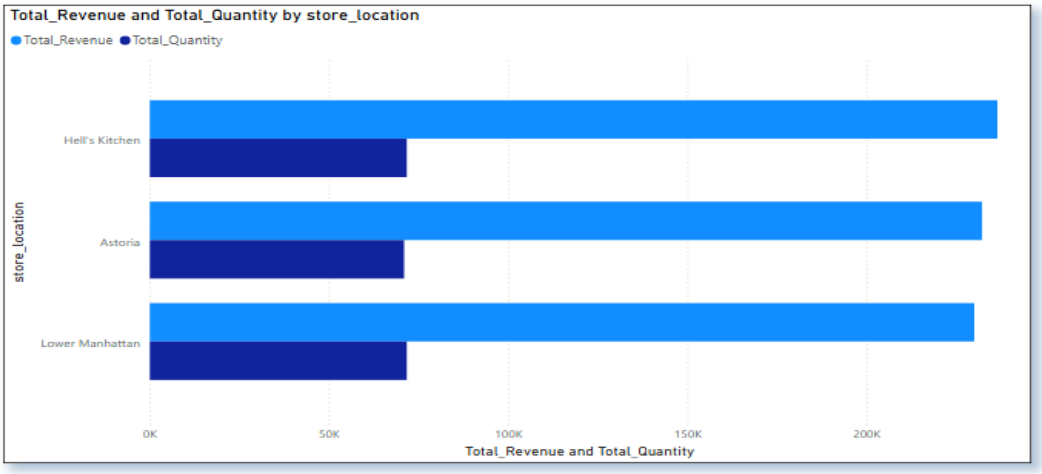
Coffee beans and Branded merchandise lead with the highest average sales value, indicating these categories command a premium price point. Items such as Loose Tea, Packaged Chocolate, and Drinking Chocolate maintain a consistent mid-range average, providing stable revenue per transaction. Daily staples like Coffee, Tea, and Bakery show lower average sales, suggesting they drive profit through high transaction frequency rather than high unit price.



10. Store-Wise Comparison of Revenue and Sales Quantity:

Insights

Hell's Kitchen leads slightly in both total revenue and item quantity, establishing itself as the flagship high-traffic location. Lower Manhattan and Astoria show nearly identical performance metrics, suggesting that the customer base and purchasing power are uniform across different urban districts. The tight correlation between revenue (light blue) and quantity (dark blue) across all stores suggests a standardized pricing model is working effectively in every region.



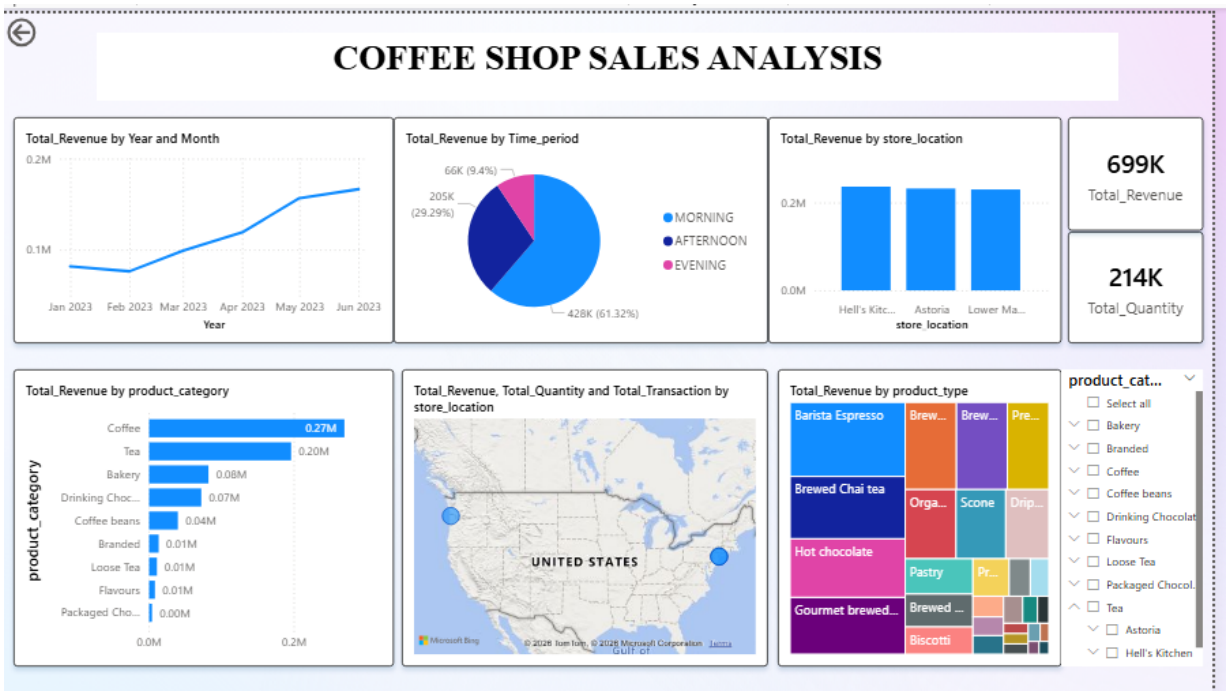
11. Transaction Volume Analysis by Order Size:

Insights

The data shows that the vast majority of orders (87,159 transactions) consist of just a single item, characterizing the business as a high-frequency "grab-and-go" model. Small-scale purchases of 1 to 2 items account for nearly 98% of the total transaction volume, highlighting the importance of speed and efficiency for individual customers.

transaction_qty	Count of transaction_id
1	87159
2	58642
3	3279
4	23
8	10
6	3
Total	**149116**

DASHBOARD



9. Insights & Conclusions

The analysis of the Coffee Shop Sales dataset provides valuable insights into sales performance, customer purchasing behavior, and product demand patterns. The sales trend over time shows fluctuations across different months, with a noticeable increase in revenue during the later months. This suggests a growing demand for coffee shop products during this period. The time-based analysis reveals that a large portion of transactions occur during the morning hours, indicating that customers frequently visit the coffee shop at the start of their day.

The analysis also identifies the products that contribute the most to total revenue. Maintaining adequate stock levels for these high-demand products can help prevent shortages and ensure customer satisfaction. Understanding these purchasing patterns enables the coffee shop management to make better decisions regarding inventory planning, product promotion, and store operations.

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10. Conclusions

The Coffee Shop Sales analysis demonstrates how Power BI can be used to transform raw transactional data into meaningful business insights.

Through data cleaning, transformation, DAX calculations, and interactive visualizations, the dashboard provides valuable insights into sales performance, product demand, and store-level performance.

These insights can help business managers make informed decisions related to inventory management, product promotion, and store operations.